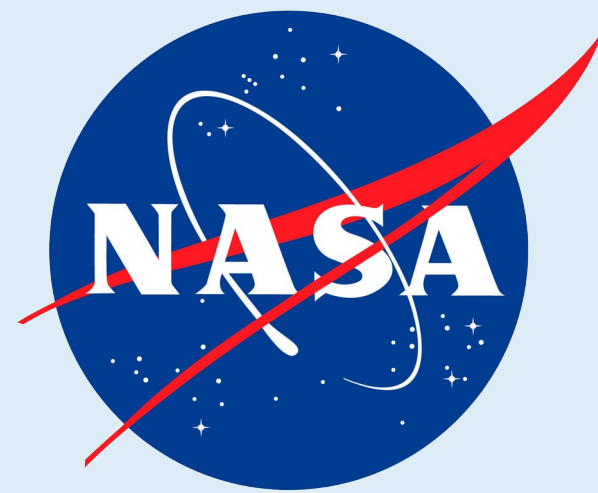




Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

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ABSTRACT

Mission differences can obscure spaceflight responses shared across studies. We mapped NASA OSDR mouse bulk RNA sequencing samples to ARCHS4 references matched by tissue. We used expiMap models constrained by Reactome to analyze about 2,000 highly variable genes. Flight versus ground shifts were balanced across projects and assessed with enrichment tests, project holdouts, three training runs, composition proxies, review of member genes, and literature annotation after scoring. Retained programs were lower for thymic repair and cytoskeletal processes, skin maintenance and barrier processes, liver adaptive immunity, and spleen adaptive and innate immunity. Spleen showed the strongest result across multiple pathways.

INTRODUCTION

PROJECT OBJECTIVE

Identify gene programs that shift consistently across spaceflight missions.

THE CENTRAL CHALLENGE

Observed expression **Spaceflight biology** + **Tissue biology** + **Animal variation** + **Study / protocol**

Study identity and protocol can dominate an unconstrained expression representation. Tissue-matched reference mapping, accession conditioning, and equal project weighting reduce this bias without claiming to remove all mission confounding.

FINAL ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples. Spleen excludes one condition-strain-confounded project.

ROBUSTNESS GATE

ssGSEA + GSEA

Held-out projects

3 full trainings

Composition proxies

Member genes

Literature review

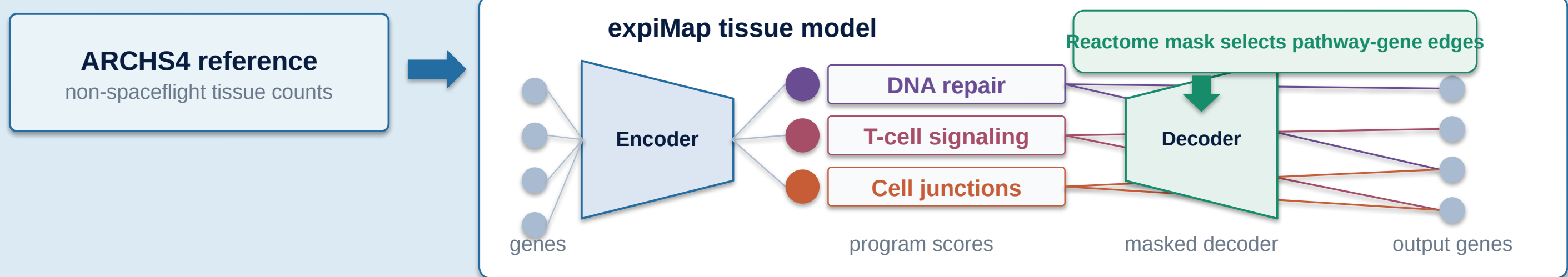
Complementary means an annotated Reactome program that adds a plausible perspective beyond the dominant phenotype in prior literature. De novo nodes were not retained in the final models.

METHODS

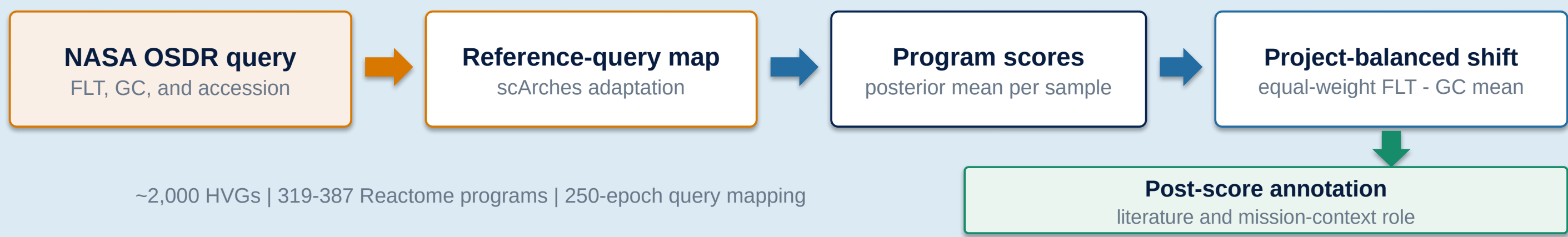
Pathway structure is wired into the decoder

Train a tissue reference, then map flight and ground samples into the same annotated program space.

1 REFERENCE TRAINING

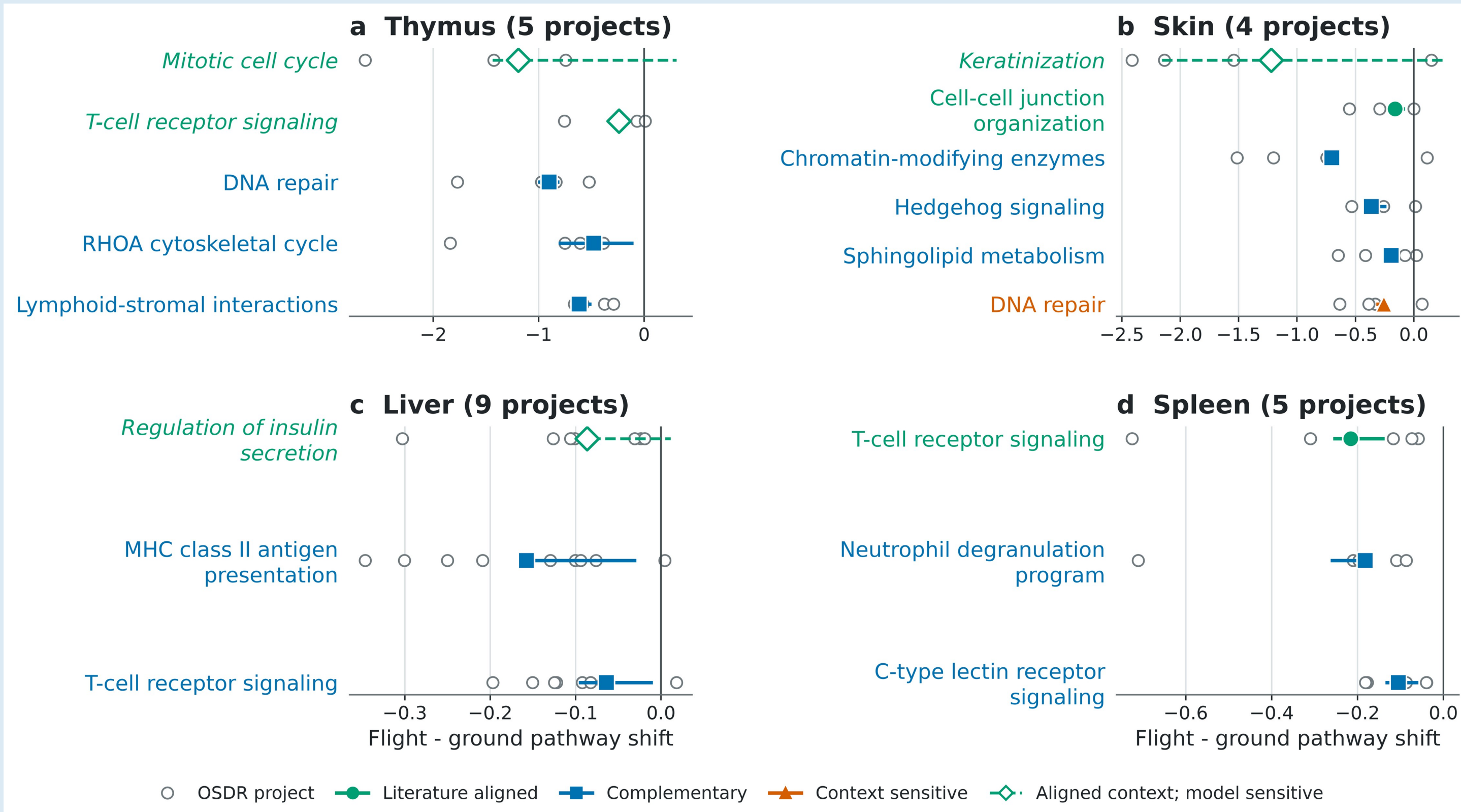


2 QUERY MAPPING AND CROSS-MISSION COMPARISON



~2,000 HVGs | 319-387 Reactome programs | 250-epoch query mapping

RESULTS: ALIGNED ANCHORS AND RETAINED PROGRAM SHIFTS



Filled markers show the 13 primary retained programs. Four open green diamonds show literature-aligned anchors that failed at least one robustness gate and provide context rather than additional claims. Colored ranges span three complete trainings; axes are tissue specific.

RESULTS: BIOLOGICAL INTERPRETATION

Final robustness-filtered tissue stories

Flight-associated pathway-score patterns and tissue-state hypotheses

Tissue context	Observed lower expiMap program scores in flight	Interpretive link (not causal order)	Tissue-state hypothesis
A THYMUS 	↓ DNA repair ↓ RHOA cytoskeletal cycle ↓ Lymphoid-stromal interactions	----->	Consistent with reduced repair, cell motility, and niche coordination
B SKIN 	↓ Chromatin-modifying enzymes ↓ DNA repair ↓ Hedgehog signaling ↓ Sphingolipid metabolism ↓ Cell-cell junction organization	----->	Consistent with reduced tissue maintenance, barrier support, and cell coordination
C LIVER 	↓ MHC class II antigen presentation ↓ T-cell receptor signaling	----->	Consistent with lower adaptive immune communication
D SPLEEN 	↓ T-cell receptor signaling ↓ Neutrophil degranulation ↓ C-type lectin receptor signaling	----->	Consistent with lower adaptive activation, innate sensing, and effector-related transcription

Conceptual synthesis of bulk-tissue pathway scores. Lower scores do not prove pathway inhibition; cell-composition shifts may contribute.

CONCLUSIONS

- Spleen was strongest: T cell receptor, neutrophil degranulation, and C-type lectin receptor programs were lower across five projects and three trainings; all had GSEA FDR <0.05.
- Skin and thymus emphasized lower tissue maintenance programs; liver showed lower adaptive immune signaling amid heterogeneous metabolism.
- Bulk latent scores are testable hypotheses, not causal or cell resolved measures of pathway activity.

REFERENCES

1 Gebre et al., NAR 2025 (OSDR). 2 Lotfollahi et al., Nat Cell Biol 2023 (expiMap). 3 Lachmann et al., Nat Commun 2018 (ARCHS4). 4 Milacic et al., NAR 2024 (Reactome). 5 Horie et al., Sci Rep 2019. 6 Cope et al., Commun Med 2024. 7 Beheshti et al., Sci Rep 2019. 8 Gridley et al., J Appl Physiol 2009.

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github.com/jasont314/nasa-mouse